Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to guide strategic business decisions

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Dataset Overview

3,900

Total Purchases

Only in Review Rating column

18

Data Columns

Comprehensive transaction data

37

Missing Values

Customer demographics & behavior

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Data Preparation Process

01

Data Loading

Imported dataset using pandas

02

Initial Exploration

Structure check and summary statistics

03

Missing Data Handling

Imputed Review Rating with median values

04

Feature Engineering

Created age groups and purchase frequency

05

Database Integration

Connected to PostgreSQL for analysis

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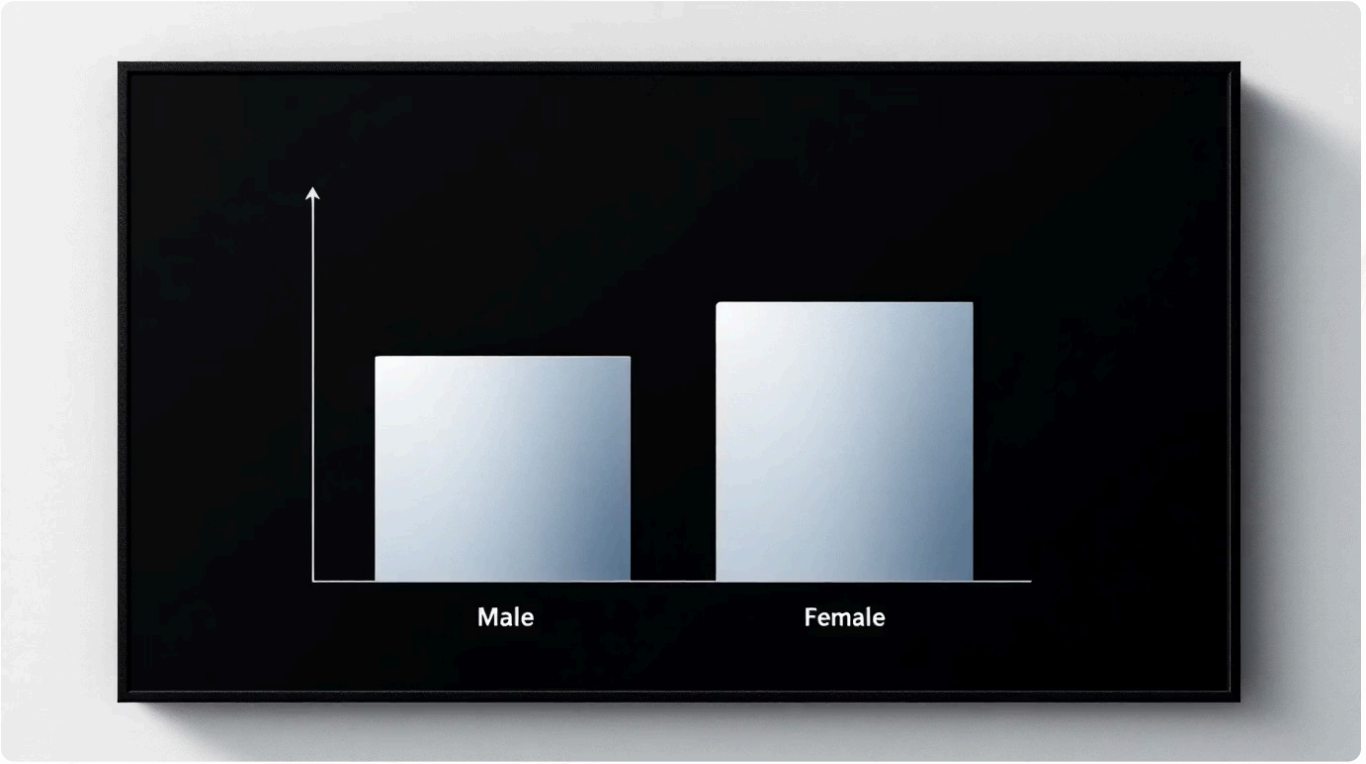
Revenue by Gender Analysis



Key Insight

Female customers generate slightly higher total revenue than male customers

Strategic Implication

Gender-based marketing strategies could optimize revenue streams

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High-Value Discount Users

Smart Shoppers

High spenders who maximize value with discounts

Strategic Opportunity

Target premium customers with exclusive offers

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Top-Rated Products

Blouse

Highest customer satisfaction

Dress

Consistently excellent reviews

Shirt

Strong customer approval

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Shipping Preferences Impact

Express Shipping

$65

Average purchase amount

Standard Shipping

$58

Average purchase amount

Express shipping customers spend 12% more per transaction

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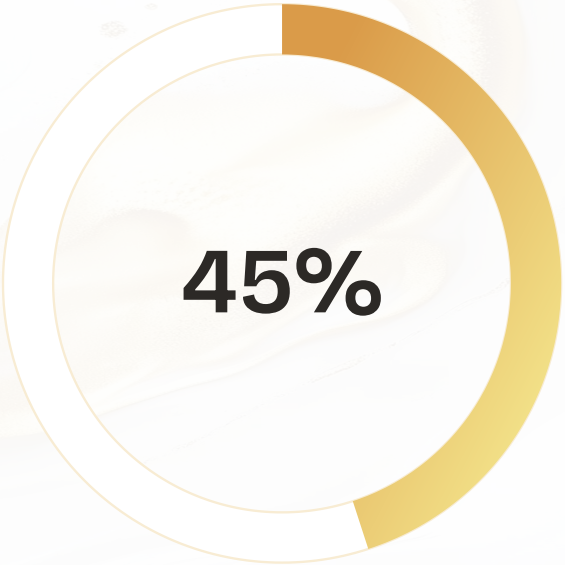
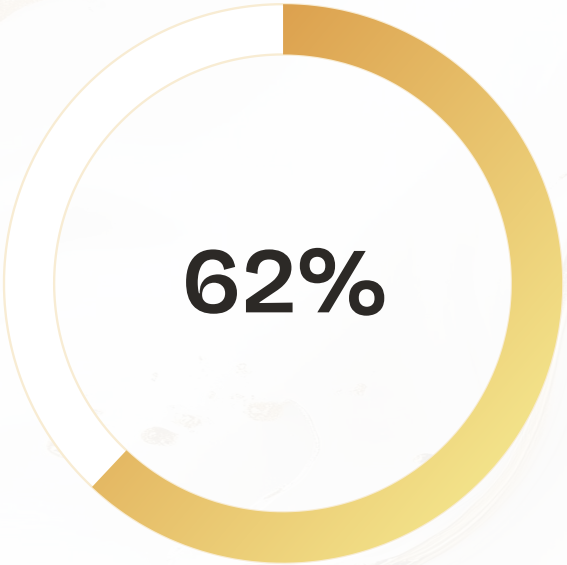
Subscription Impact



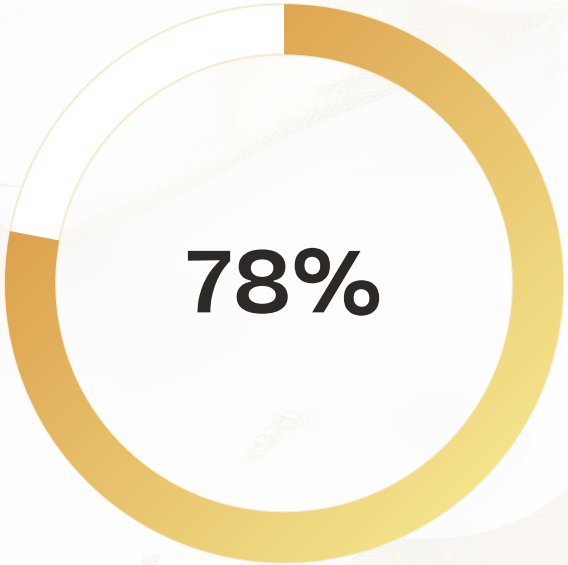
Higher Spend

Subscribers vs non-subscribers



Revenue Share

From subscription customers



Loyalty Rate

Repeat purchase frequency

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Customer Segmentation

Loyal

15% - High value customers

Returning

35% - Regular shoppers

New

50% - First-time buyers

Focus on converting New to Returning, Returning to Loyal

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Strategic Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers

Loyalty Programs

Reward repeat buyers to increase retention

Targeted Marketing

Focus on high-revenue segments and express users

Product Positioning

Highlight top-rated products in campaigns

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